EXPERIMENT 18

High Level Language Extension with Cursors

Aim:

To implement Cursors using program in MySQL.

Description:

Cursor is a Temporary Memory or Temporary Work Station. It is allocated by Database Server at the Time of Performing DML(Data Manipulation Language)operations on Table by User. Cursors are used to store Database Tables.

1. **Implicit Cursors:**

   Implicit Cursors are also known as Default Cursors of SQL SERVER. These Cursors are allocated by SQL SERVER when the user performs DML operations.

2. **Explicit Cursors:**

   Explicit Cursors are Created by Users whenever the user requires them. Explicit Cursors are used for Fetching data from Table in Row-By-Row Manner.

How to create Explicit Cursor:

1. **Declare Cursor Object.**

   Syntax : DECLARE cursor_name CURSOR FOR SELECT * FROM table_name

2. **Open Cursor Connection.**

   Syntax : OPEN cursor_connection

3. **Fetch Data from cursor.**

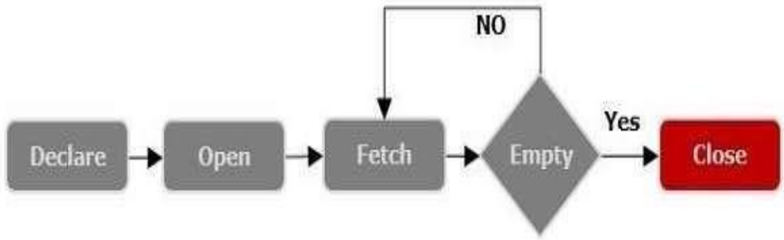
   There are total 6 methods to access data from cursor. They are as follows :
   FIRST is used to fetch only the first row from cursor table.
   LAST is used to fetch only last row from cursor table.
   NEXT is used to fetch data in forward
   direction from cursor table. **PRIOR** is used
   to fetch data in backward direction from
   cursor table.

RELATIVE n is used to fetch the data in incremental way as well as decremental way.

Syntax : FETCH NEXT/FIRST/LAST/PRIOR/ABSOLUTE n/RELATIVE n FROM cursor_name

4. **Close cursor connection.**
   Syntax : CLOSE cursor_name
5. **Deallocate cursor memory.**

   Syntax : DEALLOCATE cursor_name



Question 1

Write a Cursor program using MySQL to retrieve and build a list of patient names from the `Patient` table.

OUTPUT:

1)

```
CREATE PROCEDURE BuildPatientList(INOUT patient_list VARCHAR(4000))
BEGIN
    DECLARE v_finished INTEGER DEFAULT 0;
    DECLARE v_patient_name VARCHAR(100) DEFAULT '';

    DECLARE patient_cursor CURSOR FOR
        SELECT Patient_Name FROM Patient;

    DECLARE CONTINUE HANDLER FOR NOT FOUND
        SET v_finished = 1;

    OPEN patient_cursor;

    get_patient: LOOP

        FETCH patient_cursor INTO v_patient_name;

        IF v_finished = 1 THEN
            LEAVE get_patient;
        END IF;

        SET patient_list = CONCAT(v_patient_name, '; ', patient_list);
```

```
    END LOOP get_patient;

    CLOSE patient_cursor;

END //

DELIMITER ;

SET @patient_list = '';

CALL BuildPatientList(@patient_list);

SELECT @patient_list AS Patient_List;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Patient_List
Sneha; Rahul; Priya; Arun Kumar;

RESULT:

Thus the Cursor program using MySQL is executed successfully.